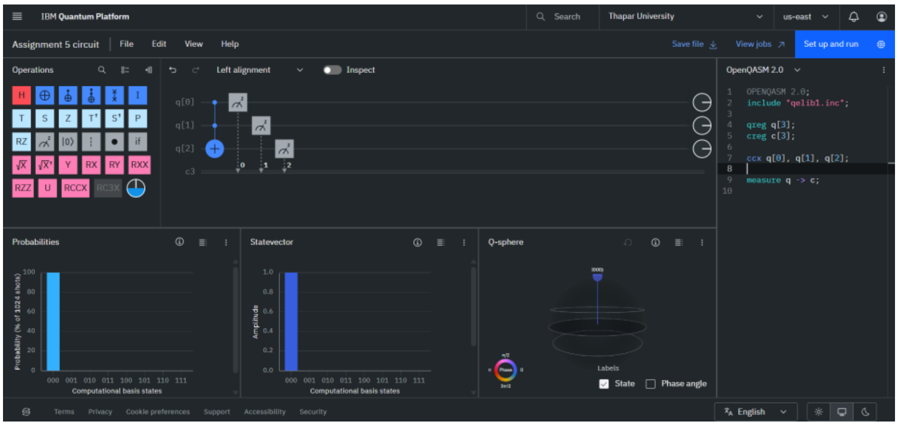
Assignment 5

Divjot Singh Gulati 102316025

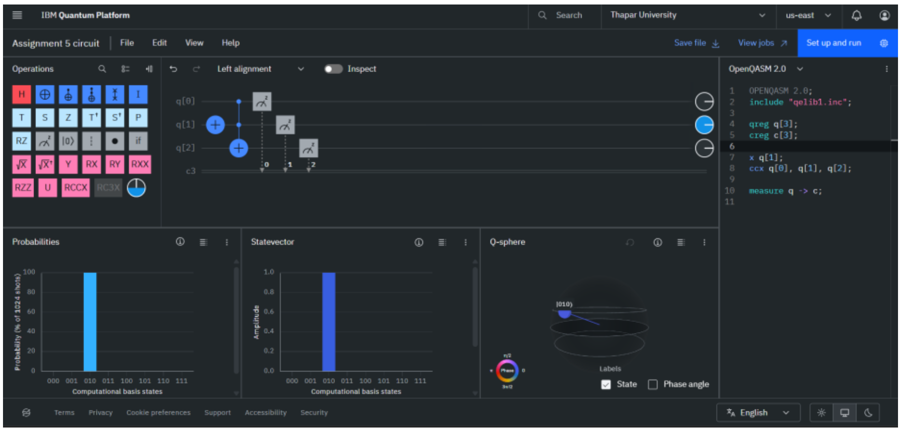
Experiment 1-:

Toffoli Gate as a Universal Classical Gate

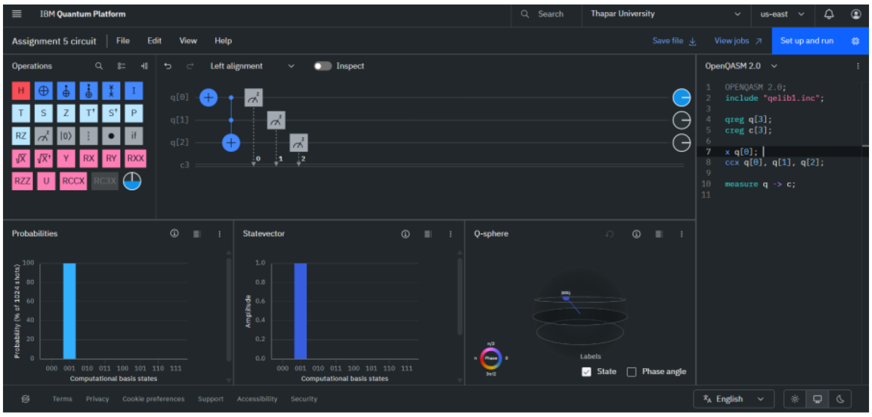
1. For AND Gate
    a. When A=0, B=0



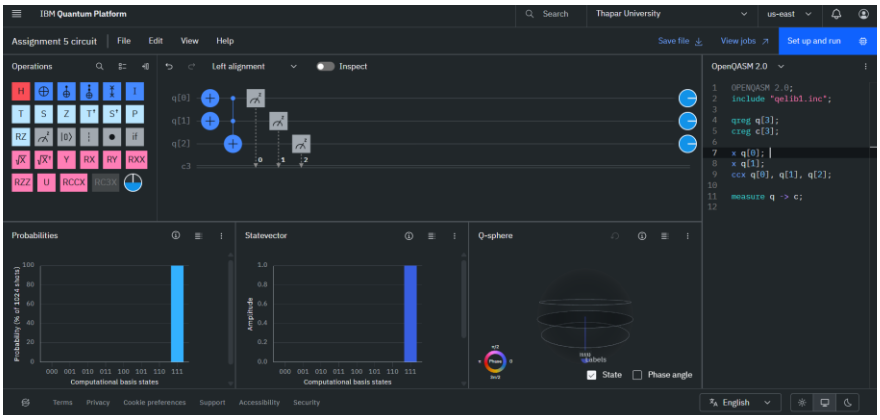
    b. When A=0, B=1



    c. When A=1, B=0

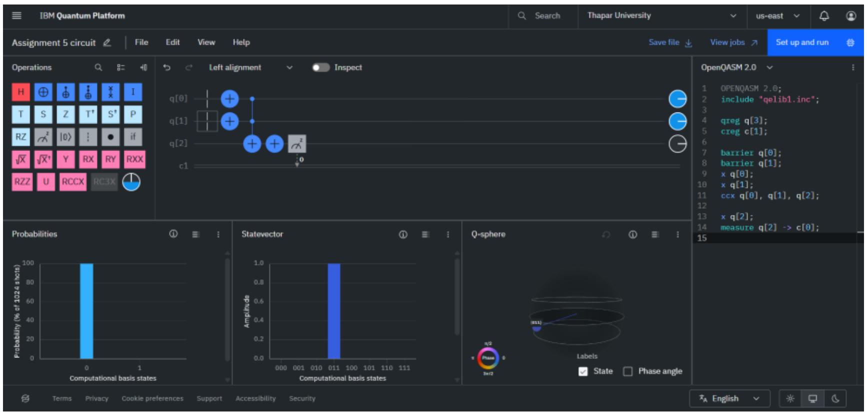


d. When A=1, B=1



2. For OR Gate

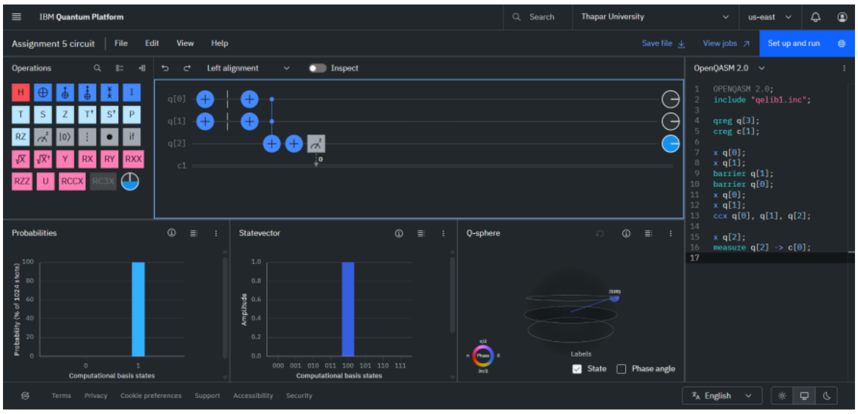
a. When A=0, B=0



b. When A=0, B=1

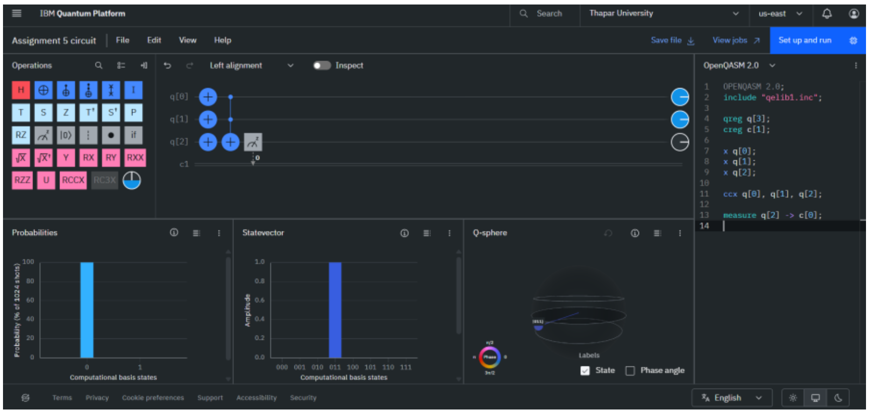
c. When A=1, B=0

d. When A=1, B=1



3. For NOT Gate
   a. When C=0, we get output as 1

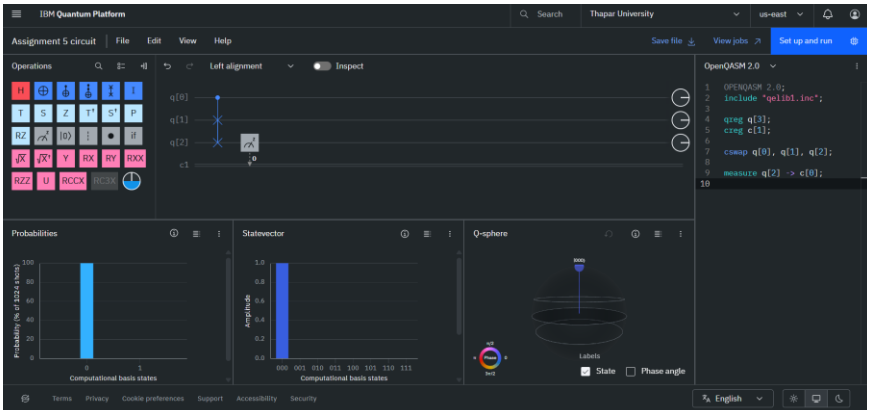
b. When C=1, we get output as 0



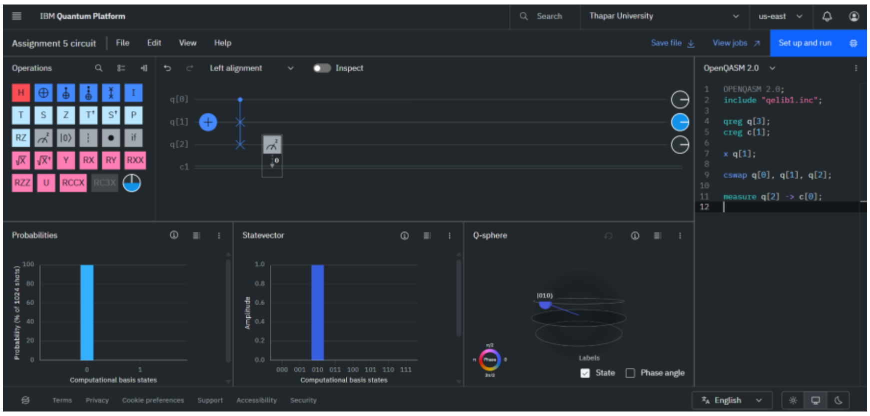
Experiment 2-:

Fredkin Gate as a Universal Classical Gate

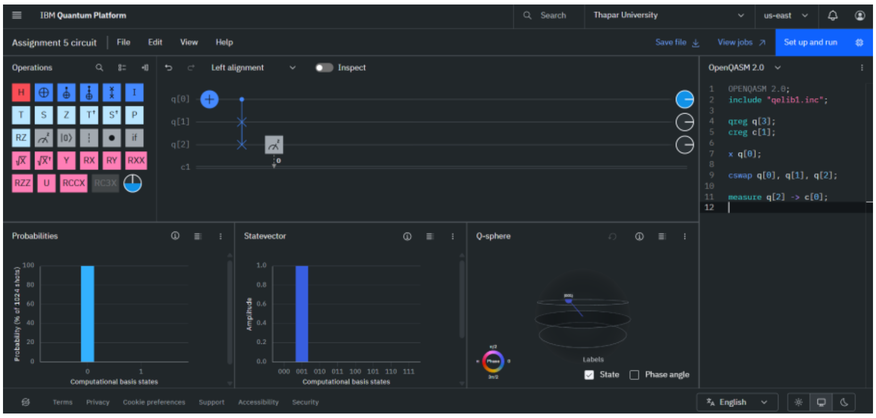
1. For AND Gate
   a. When A=0, B=0



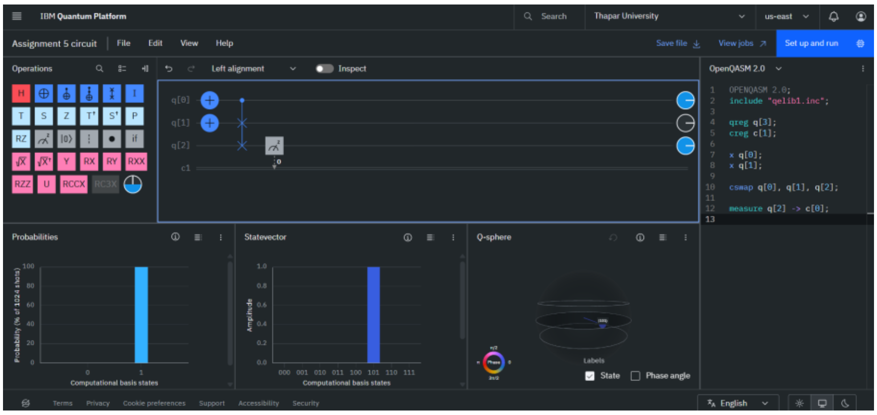
   b. When A=0, B=1



c. When A=1, B=0

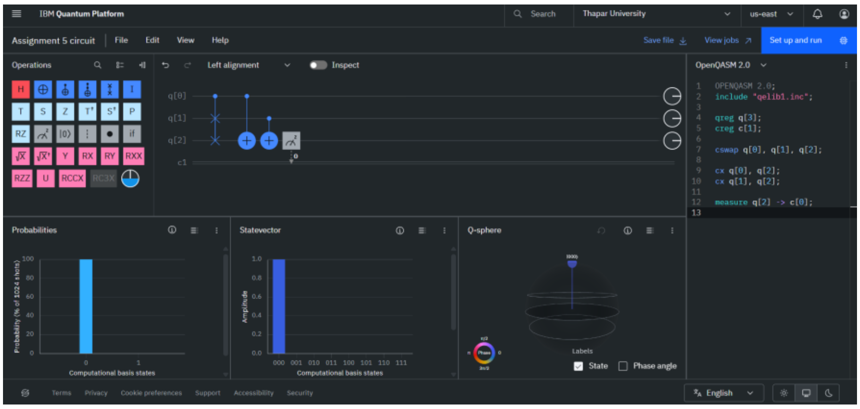


d. When A=1, B=1

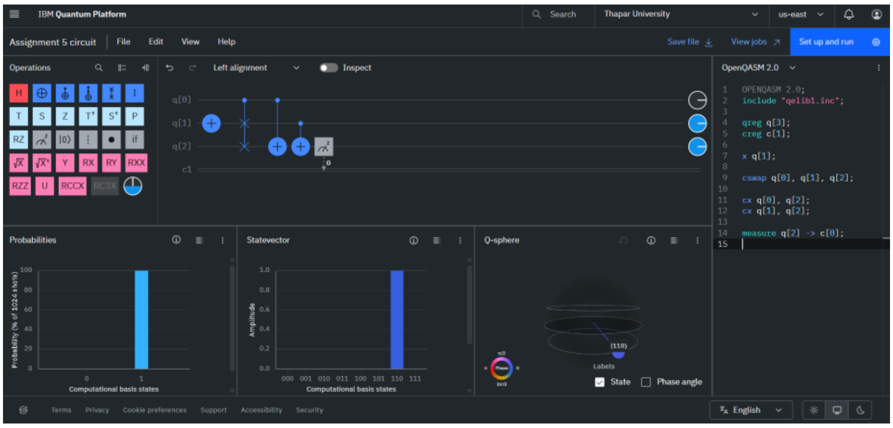


2. For OR Gate

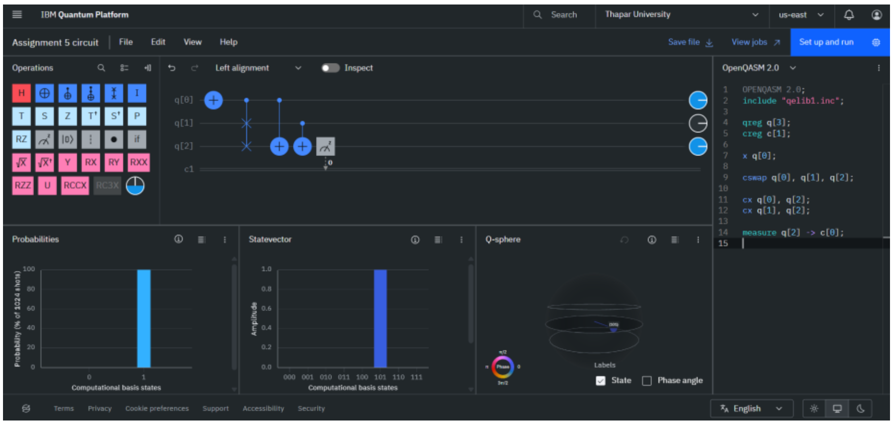
a. When A=0, B=0



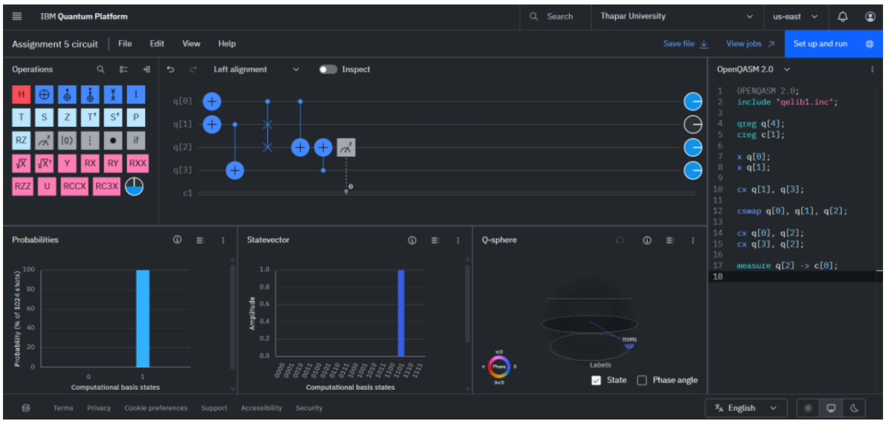
b. When A=0, B=1



c. When A=1, B=0

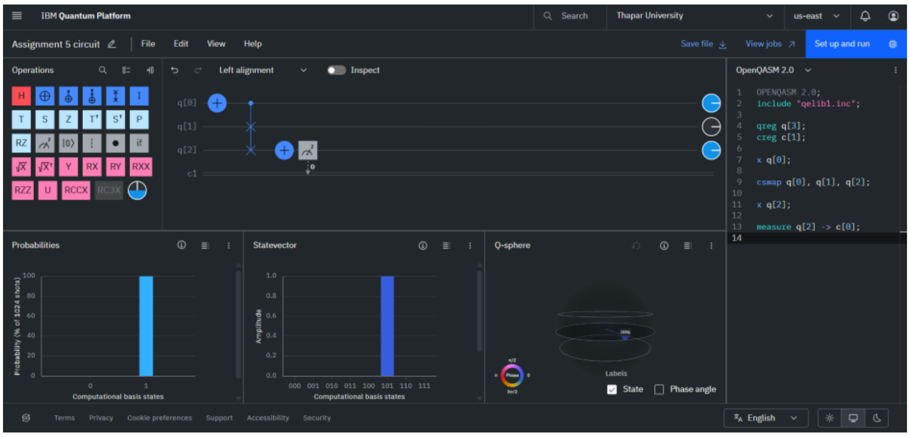


d. When A=1, B=1

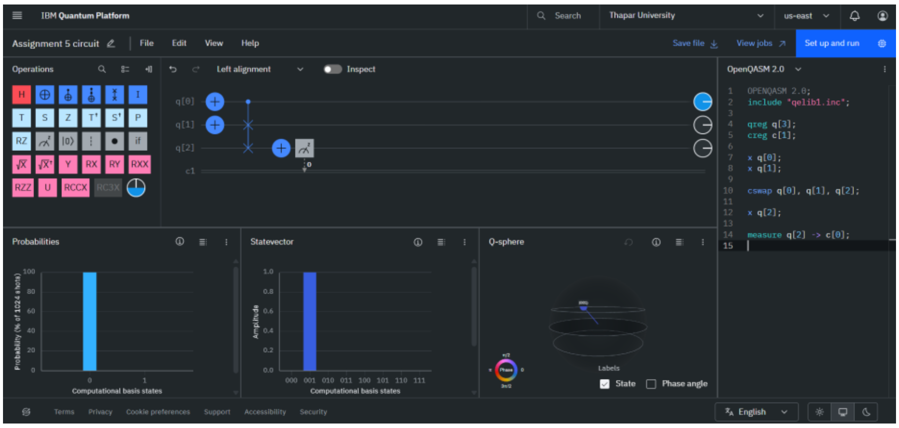


3. For NOT Gate

   a. When C=0, we get output as 1



   a. When C=1, we get output as 0

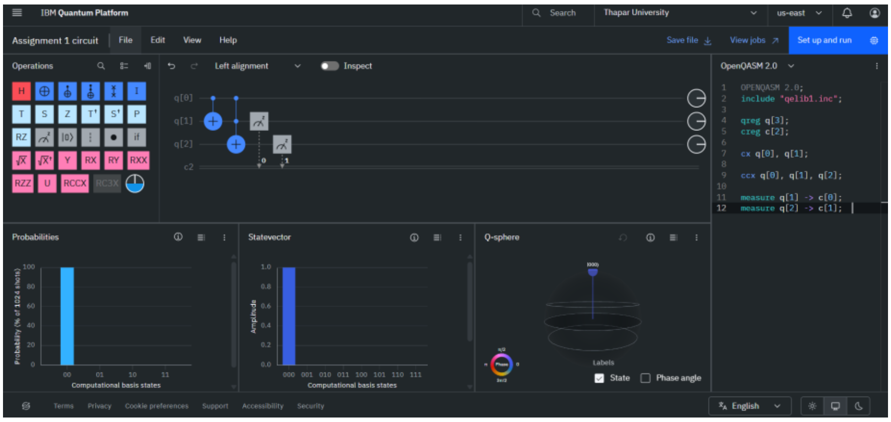


Experiment 3-:

Half and Full Adder Quantum Circuits

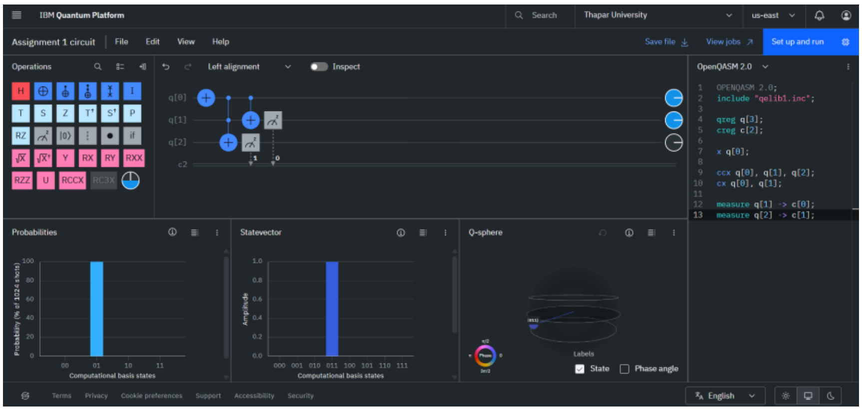
1. Half Adder

   a. When A=0, B=0



b. When A=0, B=1

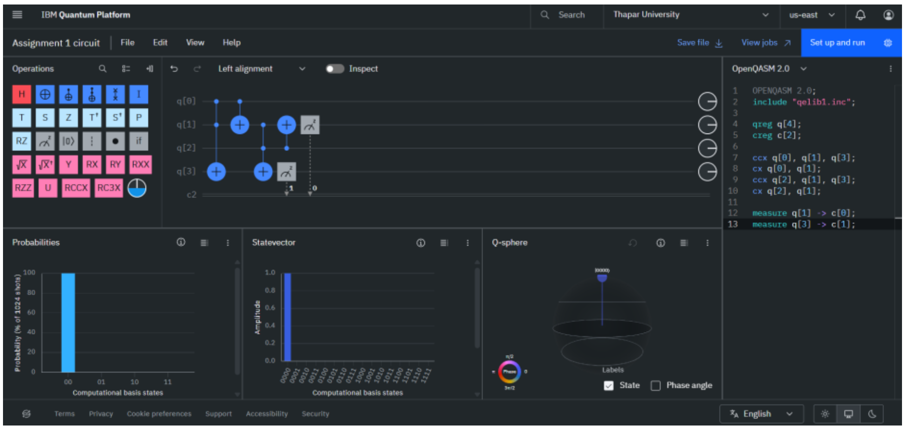
c. When A=1, B=0



d. When A=1, B=1

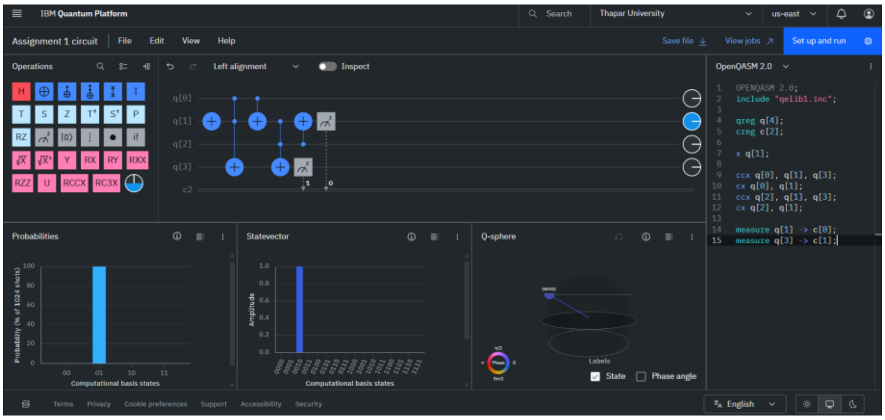
2. Full Adder

a. When A=0, B=0



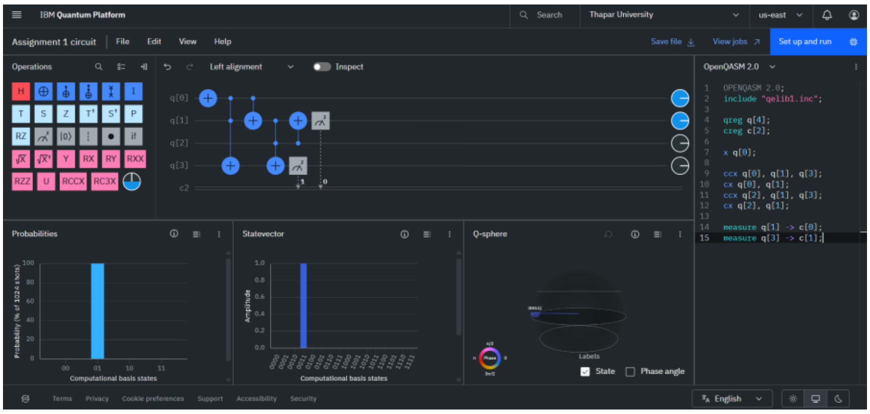
b. When A=0 B=0 Cin=1

c. When A=0 B=1 Cin=0

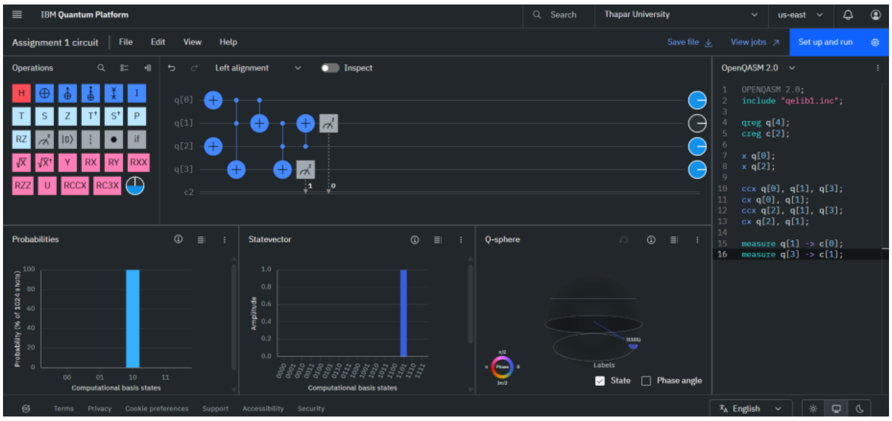


d. When A=0 B=1 Cin=1

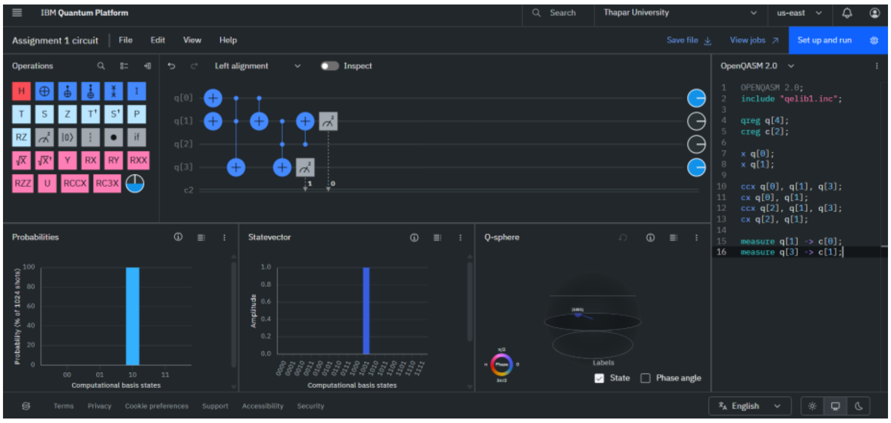
e. When A=1 B=0 Cin=0



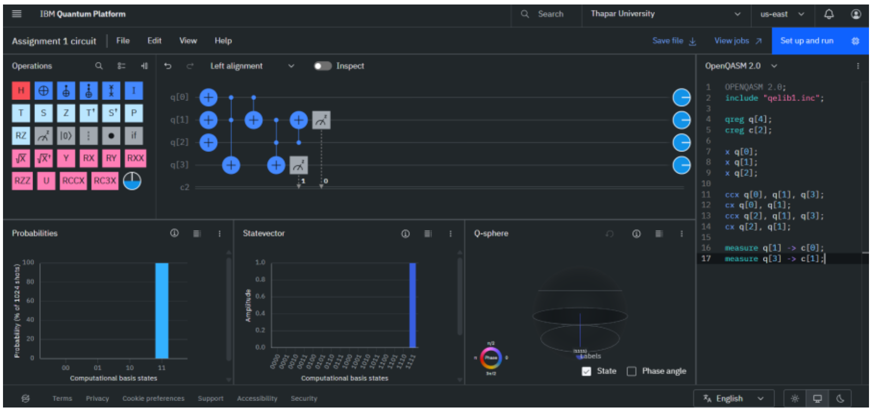
f. When A=1 B=0 Cin=1



g. When A=1 B=1 Cin=0



h. When A=1 B=1 Cin=1

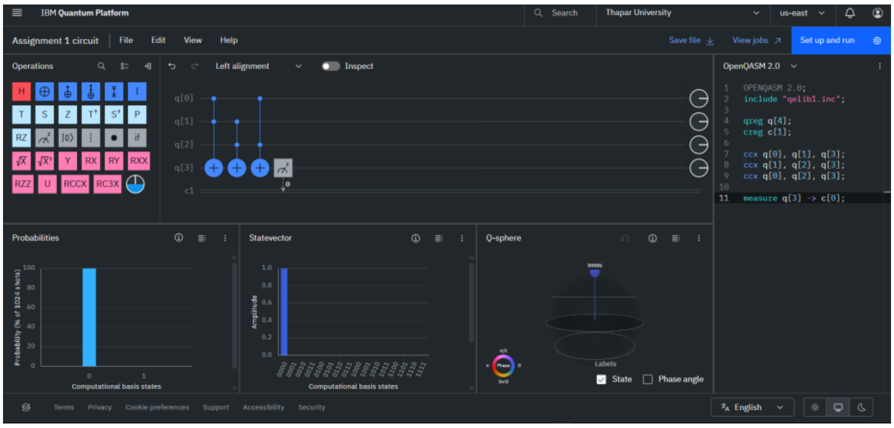


Experiment 4-:

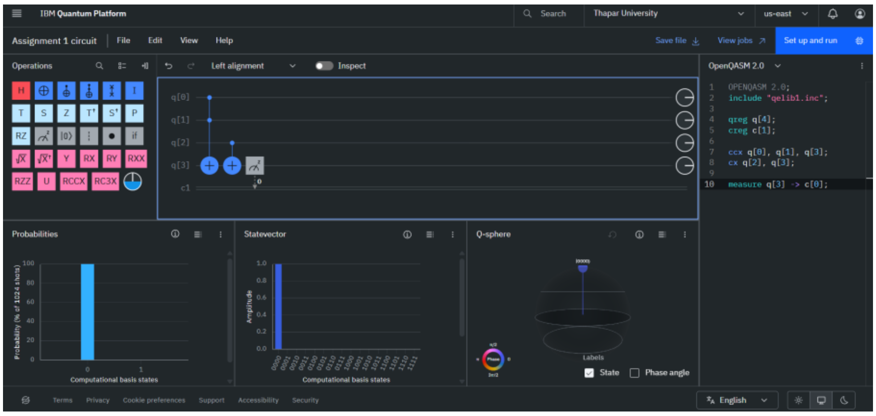
Classical Boolean Function to Quantum Circuit Conversion

1. f(x,y,z) = (x ∧y)∨z

2. f(x,y,z) = (x ∧y)∨(y ∧z)



3. f(x,y,z) = (x ∧y)⨁z



4. f(x,y,z) = (x ∧y)∨(x∧z)∨(y ∧z)